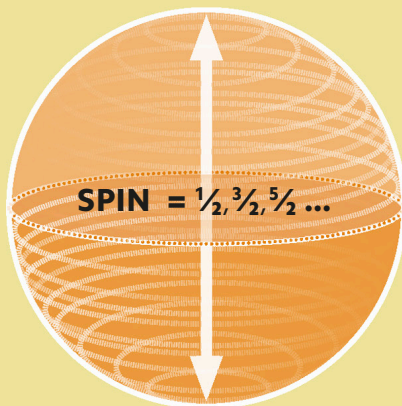


THE GREAT DIVIDE

The property of spin defines a fundamental division between elementary subatomic particles. Particles associated with the structure of matter all have spin values of either $+\frac{1}{2}$ or $-\frac{1}{2}$, while those with a spin of 1 carry the forces between these matter particles, and the spin-0 Higgs boson provides them with mass. Particles with half-integer spins follow a mathematical model called Fermi-Dirac statistics and are known as fermions. Those with integer or zero spin follow Bose-Einstein statistics and are called bosons.

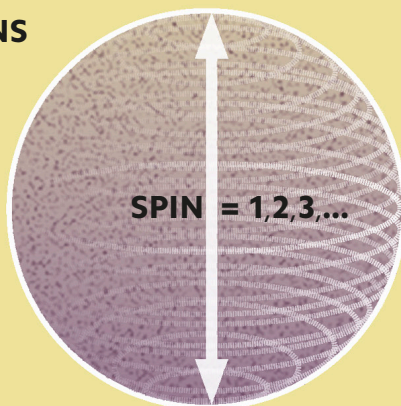
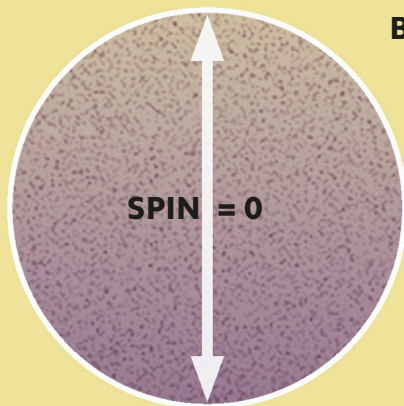
FERMIONS



Matter particles

The + or - sign on a particle's spin simply indicates whether its direction is up or down. When fermions join together their spins add up.

BOSONS



Integer spin particles

Force-carrying bosons have a spin of 1, but larger bosons can be made by adding together fermions. The unique Higgs boson has zero spin (see p.127).